
WEEK 2 INTERNSHIP DOCUMENTATION

Exploratory Data Analysis and Visualization

INTERNSHIP TRACK	Virtual Internship in Data Science using Python
TECHNICAL DOMAIN	Data Science with Python
PROJECT FOCUS	Real Estate Exploratory Data Analysis and Visualization
PRIMARY TOOLS	Python 3, Pandas, NumPy, Matplotlib, Seaborn
MILESTONE TARGET	Week 2 Submission (6-Week Virtual Internship)
SOURCE NOTEBOOK	Real Estate Data Set (1).ipynb

Technical Verification Note

All Week 2 analytical outputs, visualizations, and numerical relationships documented in this report are grounded in the supplied Jupyter Notebook. Additional grouped/aggregated analyses are explicitly calculated from the same cleaned real-estate dataset where required to complete the Week 2 deliverable.

1. Introduction

Exploratory Data Analysis (EDA) is the stage in a data science workflow where a dataset is examined systematically to understand its distributions, structure, relationships, and unusual observations. For a real-estate dataset, EDA is particularly useful for understanding how property characteristics such as area, rate per square foot, BHK count, regulatory approval, and property type relate to listing prices.

Week 2 focuses on transforming the cleaned dataset into analytical understanding through descriptive statistics, visual exploration, bivariate analysis, correlation analysis, anomaly investigation, and focused grouped aggregations. The analysis is deliberately selective so that the findings remain useful for later predictive modeling rather than becoming a collection of unrelated charts.

2. Week 2 Objectives

- Summarize the cleaned dataset and identify important numerical and categorical characteristics.
- Visualize important numerical distributions and categorical frequencies using Pandas, Matplotlib, and Seaborn.
- Study the relationships of property price with area, rate per square foot, and BHK count.
- Measure correlations among the key numerical variables.
- Identify statistical outliers and investigate potentially unusual real-estate observations.
- Use grouped averages to compare prices across meaningful BHK, RERA, and flat-type groups.
- Synthesize the EDA into evidence-based findings relevant to subsequent machine-learning preparation.

3. Dataset Overview

The Week 2 notebook continues from the cleaned Gurugram real-estate DataFrame produced in the project workflow. The final analytical dataset contains 14,223 observations and 12 standardized columns. The main numerical variables used in EDA are price, area, rate_per_sqft, and bhk_count; the categorical fields include status, property_type, locality, builder_name, rera_approval, society, company_name, and flat_type.

Metric	Verified value
Rows	14,223
Columns	12
Numerical focus	price, area, rate_per_sqft, bhk_count
Categorical focus	status, property_type, locality, builder_name, rera_approval, society, company_name, flat_type

The notebook's categorical descriptive output reports four status categories, 1,831 property-type categories, 244 localities, 709 builder names, 952 societies, 258 company names, and six flat types. This immediately highlights that some fields have very high cardinality and therefore require care during later feature engineering.

4. Initial Exploratory Analysis

The notebook uses `df.describe(include='string')` to summarize categorical fields after standardization. The executed output provides counts, unique-category counts, the most frequent category, and its frequency.

```
df.describe(include='string')
```

Field	Unique	Most frequent	Frequency
status	4	ready to move	7,823
property_type	1,831	3 bhk independent floor	392

Field	Unique	Most frequent	Frequency
locality	244	sector 65	857
builder_name	709	properties for sale in gurgaon	818
society	952	outside society	1,277
company_name	258	m3m	1,847
flat_type	6	apartment	10,894

Observation: The categorical summary shows a strongly concentrated flat-type distribution and substantial cardinality in property_type, society, builder_name, and company_name. These characteristics are important for later ML because direct one-hot encoding of every high-cardinality category could create a very large feature space.

4.1 Numerical Baseline

For Week 2 interpretation, the cleaned numerical variables were also summarized directly from the analytical DataFrame. The median area is 2,015 sq ft, while the mean is higher because of very large observations. Price is similarly right-skewed, with a median of ₹2.62 crore and a maximum of ₹122.63 crore.

Variable	Mean	Median	Minimum	Maximum
Price	₹4.00 crore	₹2.62 crore	₹0.01 crore	₹122.63 crore
Area	2,621.8	2,015.0	60	958,320
Rate per sqft	₹15,296	₹12,380	₹100	₹310,000
BHK Count	3.27	3	0	132

5. Univariate Analysis

Univariate analysis examines one variable at a time. The notebook focuses on the most useful numerical distributions and a concise set of categorical distributions.

5.1 Distribution of Property Prices

```
plt.figure(figsize=(6,4))
sns.histplot(df['price'],bins=50,kde=True)
plt.title('Distribution of Property Prices')
plt.xlabel('price')
plt.ylabel('No. of Properties')
plt.show()
```

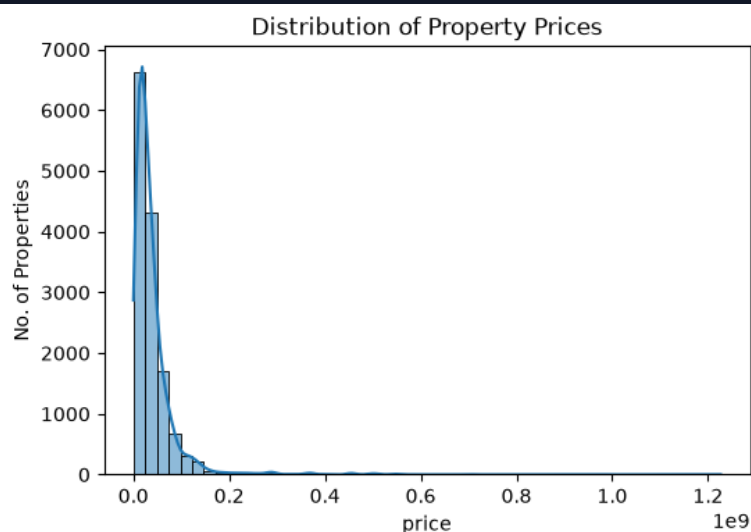


Figure 1: Distribution of Property Prices

Observation: The price distribution is strongly right-skewed. Most observations are concentrated toward the lower end of the price scale, while a smaller number of high-value properties extend far to the right. This indicates that mean-based summaries can be influenced by expensive properties and that price may require careful treatment in later predictive modeling.

5.2 Distribution of Property Area

```
sns.histplot(df[df['area']<=10000]['area'], bins=50, kde=True)
plt.title('Distribution of Property Area (Up to 10,000 sq ft)')
plt.xlabel('Area (sq ft)')
plt.ylabel('Number of Properties')
plt.xticks(range(0,10001,2000))
plt.show()
```

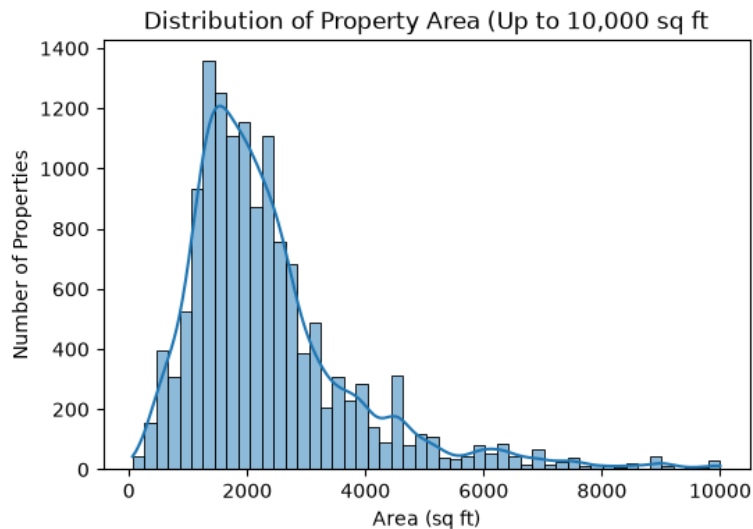


Figure 2: Distribution of Property Area — Readable View up to 10,000 sq ft

Observation: The area visualization intentionally focuses on properties up to 10,000 sq ft so that the typical residential distribution remains readable. This is a visualization filter only; the underlying dataset is not modified by this plotting step.

5.3 Distribution of BHK Count

```
sns.countplot(x='bhk_count', data=df)
plt.title('Distribution of BHK Count')
plt.xlabel('BHK Count')
plt.ylabel('Number of Properties')
plt.xticks(rotation=45)
plt.show()
```

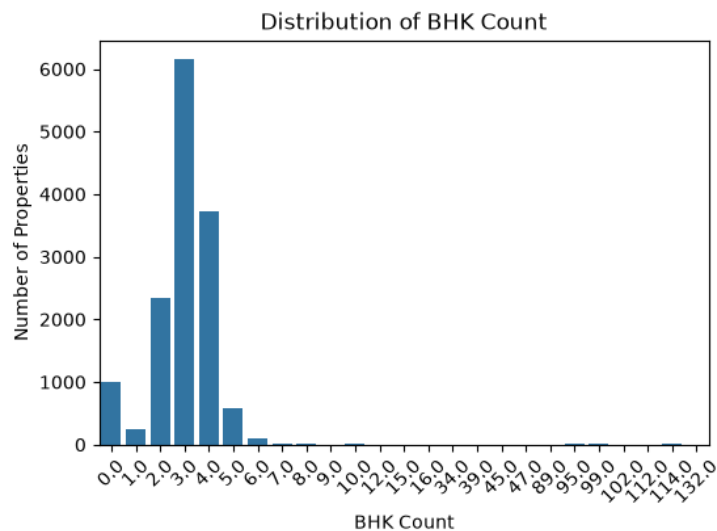


Figure 3: Distribution of BHK Count

Observation: The distribution is concentrated around common residential configurations, especially 2–4 BHK properties. The presence of unusually high BHK values requires contextual interpretation rather than automatic deletion.

5.4 Categorical Distributions

```
df['flat_type'].value_counts().plot(kind='bar')
plt.title('Distribution of Flat Type')
plt.xlabel('flat type')
plt.ylabel('No. of Properties')
plt.show()
```

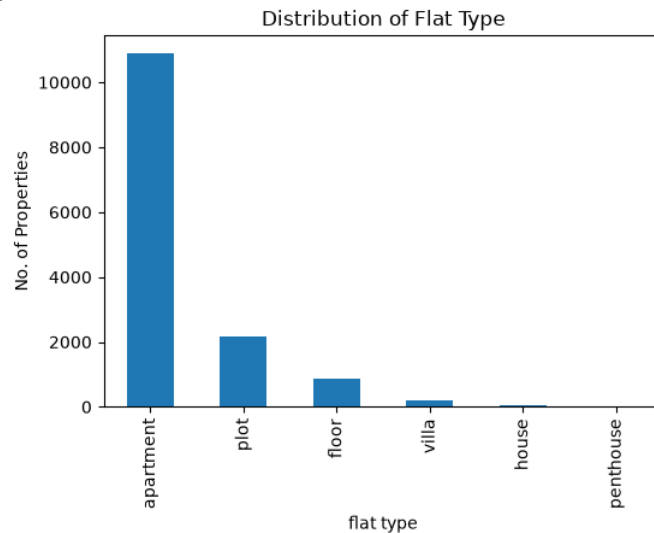


Figure 4: Distribution of Flat Type

Observation: Apartments dominate the inventory by a wide margin, followed by plots and floors. Villas, houses, and penthouses occur much less frequently. This imbalance should be considered when evaluating categorical features for modeling.

5.5 Property Status

```
status_counts = df['status'].value_counts()
plt.pie(status_counts, labels=None, autopct='%1.1f%%', startangle=90, pctdistance=1.15)
plt.title('Property Status Distribution')
plt.legend(status_counts.index, title='Property Status', loc='center left', bbox_to_anchor=(1, 0.5))
plt.tight_layout()
plt.show()
```

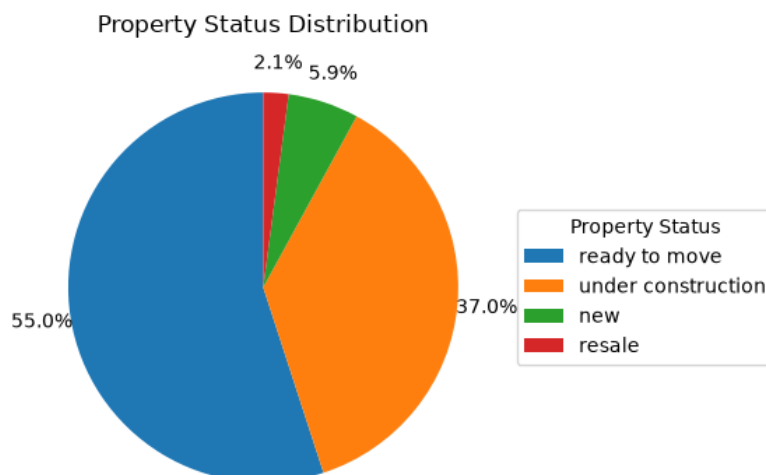


Figure 5: Property Status Distribution

Observation: The executed distribution is dominated by ready-to-move listings, followed by under-construction properties. New and resale listings form smaller groups. Status therefore provides meaningful market-context information and may be useful as a categorical predictor.

5.6 RERA Approval

Rera Approval Distribution

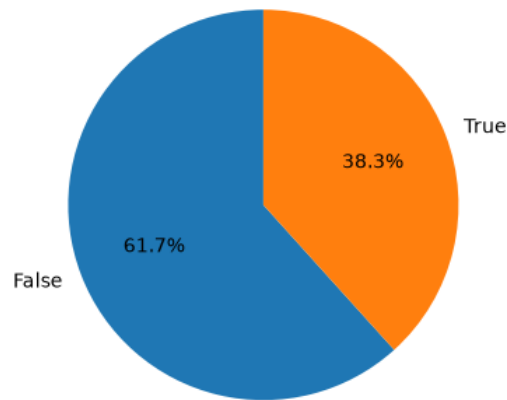


Figure 6: RERA Approval Distribution

Observation: Both RERA-approved and non-approved listings are represented in the dataset. This makes rera_approval suitable for direct grouped comparison and, after appropriate encoding, a potential model feature.

5.7 Top Companies by Number of Properties

```
top_companies = df['company_name'].value_counts().head(10)
top_companies.plot(kind='bar')
plt.title('Top 10 Companies by Number of Properties')
plt.xlabel('Company Name')
plt.ylabel('Number of Properties')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

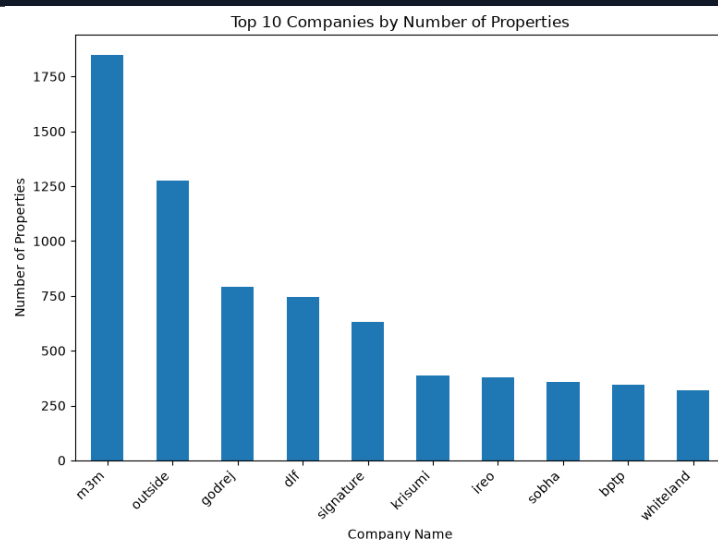


Figure 7: Top 10 Companies by Number of Properties

Observation: A relatively small set of companies accounts for a substantial share of the listings. This indicates concentration in the dataset and suggests that company_name may carry useful information, although its categorical representation should be handled carefully.

6. Bivariate Analysis

Bivariate analysis examines the relationship between property price and the three numerical variables most directly relevant to a future price-prediction task.

6.1 Price vs Area

```
sns.scatterplot(x='area', y='price', data=df)
plt.title('Price vs Area')
plt.xlabel('area')
plt.ylabel('price')
plt.tight_layout()
plt.show()
```

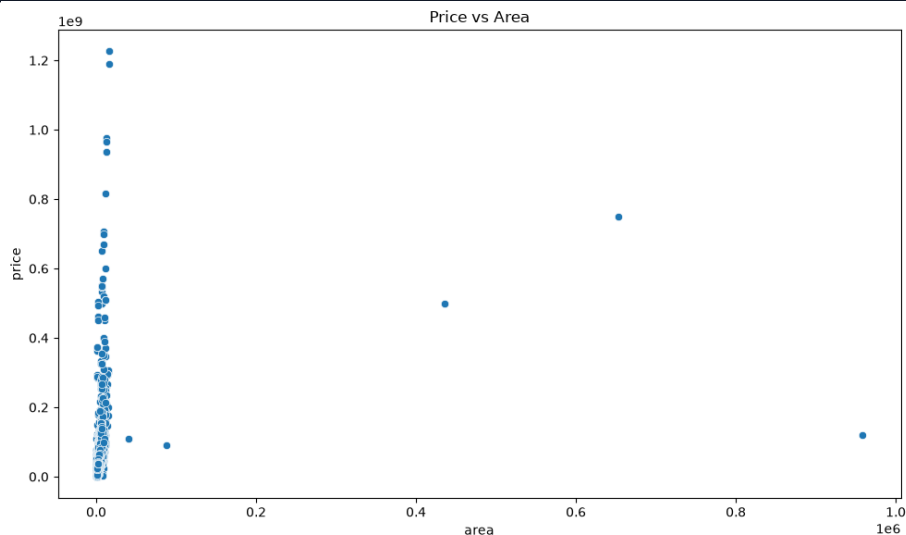


Figure 8: Price vs Area

Observation: The scatter plot shows a positive but relatively dispersed relationship. Larger properties tend to be associated with higher prices, but area alone does not explain the full variation in price. The wide spread indicates that other characteristics, especially unit rate and property context, also matter.

6.2 Property Price vs Rate per Sqft

```
sns.scatterplot(x='rate_per_sqft', y='price', data=df)
plt.title('Property Price vs Rate per Sqft')
plt.xlabel('Rate per Sqft')
plt.ylabel('Price')
plt.tight_layout()
plt.show()
```



Figure 9: Property Price vs Rate per Sqft

Observation: The relationship is visibly stronger than the price-area relationship. Higher unit rates generally coincide with higher total property prices, although the spread remains substantial.

6.3 Property Price vs BHK Count

```
sns.scatterplot(x='bhk_count', y='price', data=df)
plt.title('Property Price vs BHK Count')
plt.xlabel('BHK Count')
plt.ylabel('Price')
plt.tight_layout()
plt.show()
```

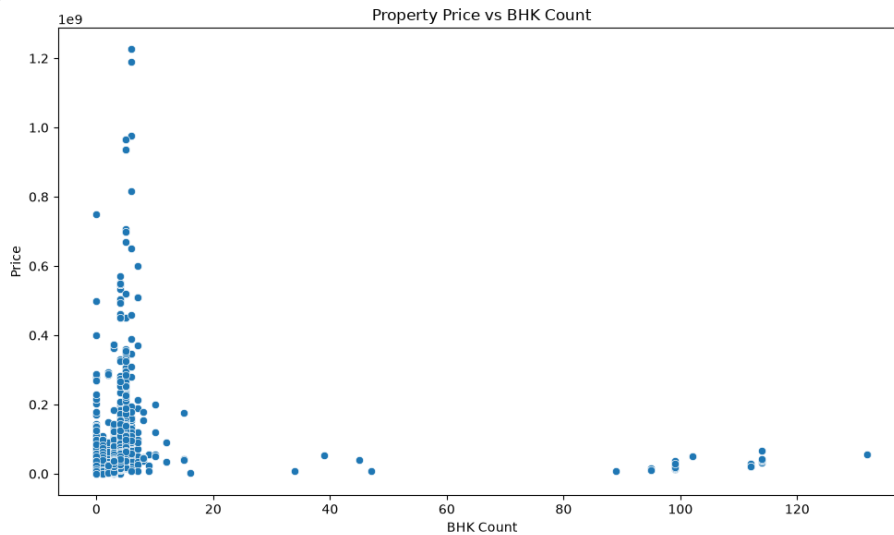


Figure 10: Property Price vs BHK Count

Observation: The relationship between BHK count and price is much weaker than the relationship with rate per square foot. Common BHK groups show a broad range of prices, indicating that BHK count alone is not sufficient to explain property value.

7. Correlation Analysis

Pearson correlation was calculated for the four key numerical variables. The resulting matrix is reproduced below from the executed analysis, followed by the notebook heatmap.

```
correlation_matrix=df[['price', 'area', 'rate_per_sqft', 'bhk_count']].corr()
correlation_matrix
```

	price	area	rate_per_sqft	bhk_count
price	1.0000	0.1974	0.6153	0.0581
area	0.1974	1.0000	-0.0008	0.0101
rate_per_sqft	0.6153	-0.0008	1.0000	0.0072
bhk_count	0.0581	0.0101	0.0072	1.0000

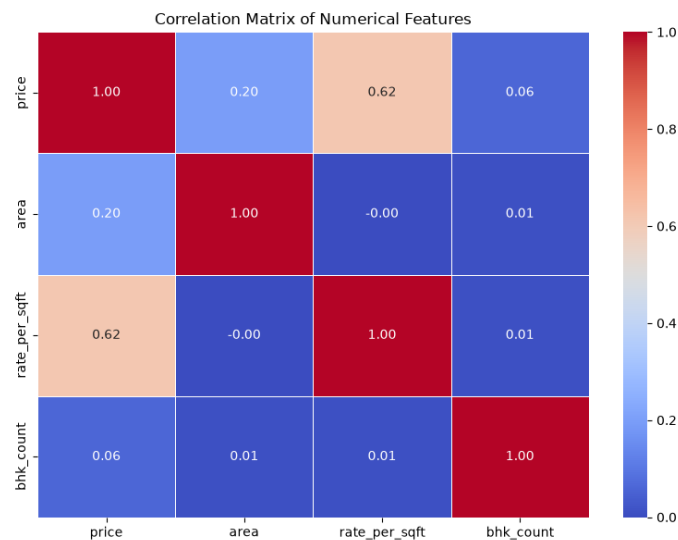


Figure 11: Correlation Matrix of Numerical Features

Key interpretation: rate_per_sqft has the strongest linear association with price ($r = 0.615$), while area has a weaker positive association ($r = 0.197$). BHK count has a very weak linear association with price ($r = 0.058$). The predictor variables are also almost uncorrelated with one another in this simple Pearson analysis.

8. Outlier & Anomaly Analysis

The notebook uses boxplots to identify extreme observations in price, area, and rate per square foot. The objective is to identify values requiring investigation, not to automatically remove them.

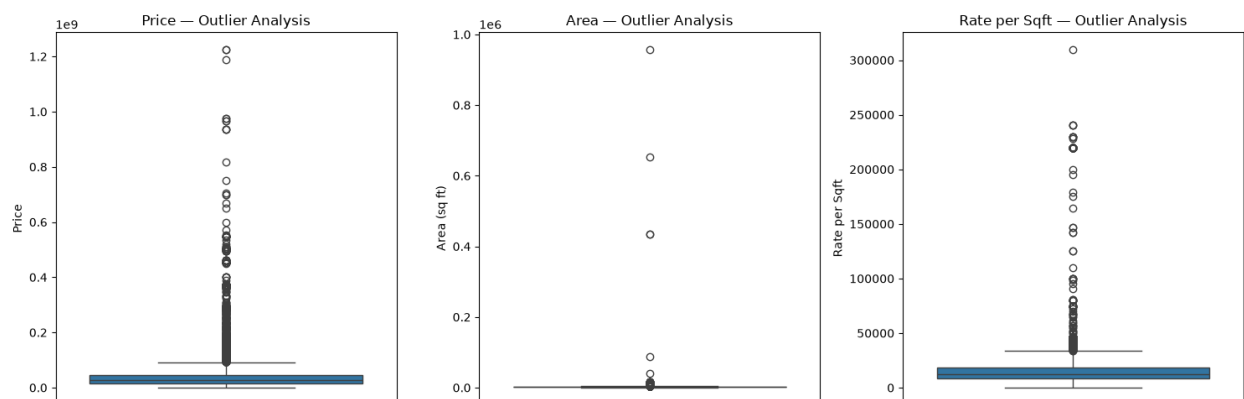


Figure 12: Price, Area, and Rate per Sqft — Outlier Analysis

The boxplots show clear upper-tail observations in all three numerical variables. Using the conventional $1.5 \times \text{IQR}$ rule on the cleaned analytical dataset, approximately 6.8% of prices, 6.7% of areas, and 3.3% of rate_per_sqft observations fall beyond the corresponding upper/lower fences. These are statistical outliers, not proof of erroneous records.

8.1 Extreme Price Observations

Price	Area	Rate/sqft	BHK	Locality
■122.63 crore	16,500	■74,323	6	Sector 42
■122.63 crore	16,500	■74,323	6	Sector 42
■118.91 crore	16,000	■74,324	6	Sector 42
■97.68 crore	13,143	■74,323	6	Sector 42
■97.68 crore	13,143	■74,323	6	Sector 42
■96.62 crore	13,000	■74,323	5	Sector 42

Observation: The highest-priced observations include luxury properties in DLF Camellias and other premium developments. Their presence is consistent with genuine high-end real-estate inventory, so they should be investigated rather than indiscriminately removed.

8.2 Extreme Area Observations

Area	Price	Rate/sqft	BHK	Locality
958,320	■11.97 crore	■125	0	37D Dwarka Exp Gurgaon
653,400	■75.00 crore	■1,147	0	Baliawas
435,600	■50.00 crore	■1,147	0	Baliawas
435,600	■50.00 crore	■1,147	0	Baliawas
87,300	■9.00 crore	■1,030	0	Sector 75
40,500	■11.00 crore	■2,716	0	Sector 54

Observation: The largest-area records include residential plots and other unusual property entries. This explains why the overall area distribution is highly skewed and reinforces the need to distinguish statistical extremes from data errors.

8.3 BHK Anomaly Context

The dataset contains BHK_Count values as high as 132. These observations were retained rather than deleted because a high numeric value alone is insufficient evidence of an invalid listing; the underlying inventory can include unusual property structures or source-level mappings. Any future ML pipeline should consider a contextual feature-engineering strategy rather than blindly discarding these records.

9. Grouped / Aggregated Analysis

Grouped analysis converts the raw EDA into comparisons of average property prices across meaningful categories. These calculations use the same cleaned analytical dataset and are intentionally limited to three high-value comparisons.

9.1 Average Price by BHK Count

```
bhk_price = (
df[df['bhk_count'] <= 10]
.groupby('bhk_count')['price']
.mean()
)
```

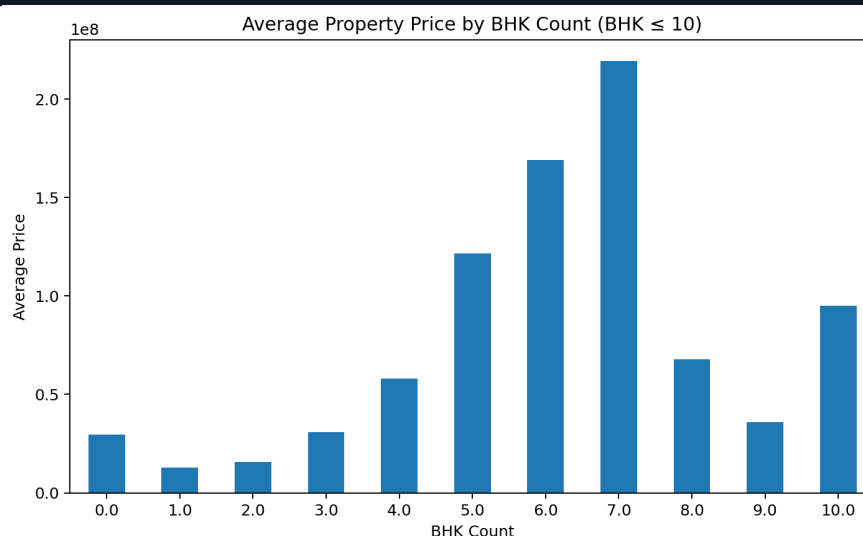


Figure 13: Average Property Price by BHK Count (BHK ≤ 10)

BHK	Average price
0	■2.97 crore
1	■1.29 crore
2	■1.57 crore
3	■3.08 crore
4	■5.82 crore
5	■12.17 crore
6	■16.91 crore
7	■21.93 crore
8	■6.80 crore
9	■3.58 crore
10	■9.50 crore

Observation: Average prices generally rise across the common 2–6 BHK range, although the trend is not perfectly monotonic at the extremes. The grouped result shows that BHK count can still be informative even though its simple Pearson correlation with price is weak.

9.2 Average Price by RERA Approval

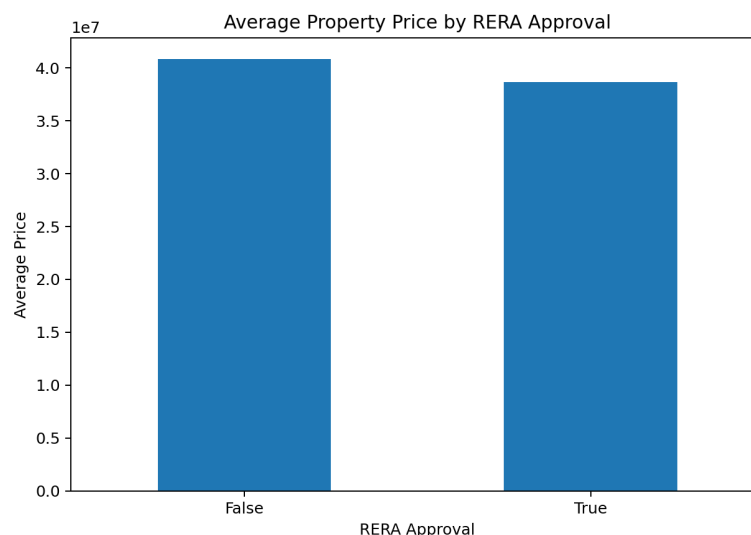


Figure 14: Average Property Price by RERA Approval

RERA Approval	Listings	Average price	Median price
Not approved	8,774	■4.08 crore	■2.58 crore
Approved	5,449	■3.86 crore	■2.71 crore

Observation: The mean price is somewhat higher for non-approved listings in this dataset, while the median is higher for approved listings. This contrast shows why both mean and median can be informative and why RERA status should not be interpreted causally from a simple grouped average.

9.3 Average Price by Flat Type

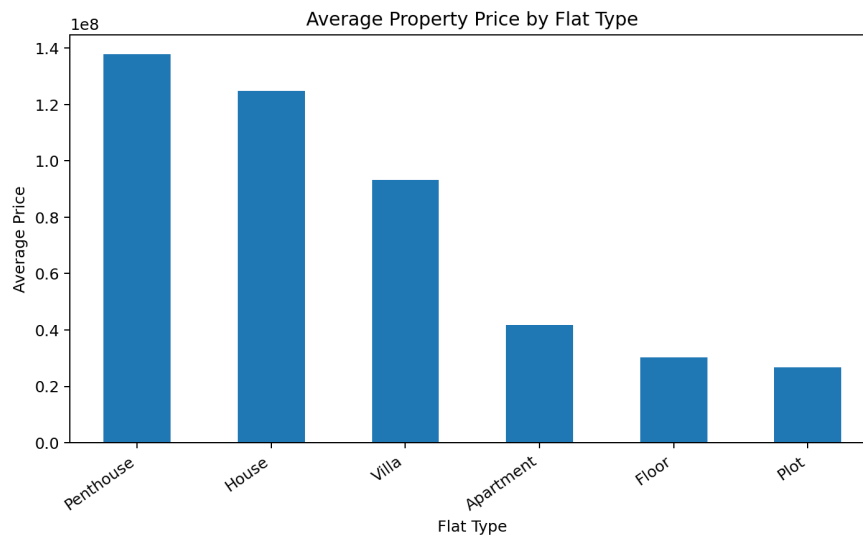


Figure 15: Average Property Price by Flat Type

Flat Type	Listings	Average price	Median price
Penthouse	32	■13.78 crore	■9.50 crore
House	45	■12.47 crore	■10.10 crore
Villa	205	■9.33 crore	■8.26 crore
Apartment	10,894	■4.17 crore	■2.80 crore
Floor	881	■3.03 crore	■2.30 crore
Plot	2,166	■2.67 crore	■1.67 crore

Observation: Penthouses, houses, and villas have the highest average prices, while apartments, floors, and plots have lower average values. The result supports the importance of property structure/type as a potentially useful categorical feature.

10. Key Findings & Insights

- Property prices are strongly right-skewed.** Most listings cluster toward the lower end of the price scale, while a smaller number of premium properties extend the upper tail. This makes robust summaries and careful target preprocessing important for later modeling.
- Rate per square foot is the strongest simple numerical correlate of price.** The Pearson correlation is approximately 0.615, stronger than the price-area correlation. Unit pricing therefore carries substantial information about total property value.
- Area has a positive but weak-to-moderate linear relationship with price.** The correlation is approximately 0.197. Larger properties generally cost more, but the wide scatter indicates that location, property type, and unit rate contribute substantially to price differences.
- BHK count has very weak linear correlation with price.** The correlation is approximately 0.058. Nevertheless, grouped averages show meaningful price differences among common BHK categories, indicating that the relationship may be nonlinear and influenced by other property characteristics.
- The dataset contains substantial statistical outliers.** The IQR analysis identifies upper-tail observations in price, area, and rate per square foot. Extreme values include genuine premium properties and very large residential plots, so statistical outlier status should not be equated with invalid data.
- Property structure is strongly associated with average price.** The grouped flat-type analysis shows much higher average prices for penthouses, houses, and villas than for apartments, floors, and plots. This suggests that categorical structure can add information beyond numerical size variables.

7. RERA approval does not show a simple one-directional price effect. The mean price is higher for non-approved listings in this dataset, while the median is higher for approved listings. The difference demonstrates the influence of distribution shape and outliers and should not be treated as a causal conclusion.

8. Several categorical variables have high cardinality. The notebook reports 1,831 property-type categories, 952 societies, 709 builders, and 258 companies. These fields may be useful, but they will require thoughtful encoding or feature engineering rather than naive one-hot encoding of every category.

9. The dataset is highly concentrated in apartments and ready-to-move listings. Apartment listings dominate flat types, and ready-to-move properties form the largest status group. This concentration should be considered when interpreting category-based patterns and future model performance.

11. Conclusion

The Week 2 exploratory analysis provides a focused understanding of the Gurugram real-estate dataset through statistical summaries, distributions, categorical comparisons, bivariate relationships, correlation analysis, outlier investigation, and grouped price comparisons. The analysis shows that property price is highly skewed and that rate per square foot has the strongest simple linear association with price among the key numerical variables examined.

Area contributes positively but does not explain the full variation in price, while BHK count has weak linear correlation despite showing meaningful differences in grouped average prices. The anomaly analysis identifies substantial upper-tail observations in price, area, and unit rate, including records that appear consistent with premium properties or large plots. These findings support careful treatment of extreme observations rather than automatic deletion.

The categorical analysis also reveals strong concentration in apartments and ready-to-move listings and high cardinality in fields such as property_type, society, builder_name, and company_name. Overall, the EDA establishes the main patterns, limitations, and feature considerations that should guide the next stage of machine-learning preparation and eventual property-price prediction.

12. Requirement-to-Evidence Mapping

Internship requirement	Work performed	Notebook / evidence	Result
Initial analysis	Categorical descriptive statistics and numerical baseline	Week 2 EDA section; <code>df.describe(include='string')</code>	Category counts, frequency concentration, numerical summary
Multiple visualizations	Numerical and categorical distributions	Executed PNG outputs: Figures 1–7	Distributions and category concentration identified
Bivariate analysis	Price vs area, rate/sqft, BHK	Executed scatter plots: Figures 8–10	Relationships and dispersion examined
Correlation	Pearson matrix + heatmap	Executed matrix + heatmap: Figure 11	Rate/sqft strongest simple correlation with price
Anomaly analysis	Boxplots + extreme-record inspection	Executed boxplots + top-value tables: Figure 12	Statistical outliers contextualized
Grouped analysis	Average price by BHK, RERA, flat type	Calculated from cleaned dataset: Figures 13–15	Category-level price differences quantified
Interpretation	Key findings and conclusion	Sections 10–11	Evidence-based insights synthesized

Final verification: This report contains no screenshot placeholders, unfinished figure slots, or manual-insertion instructions. The visual outputs from the notebook are embedded directly, and the additional grouped analyses are calculated from the supplied cleaned dataset.